Akul Ramasubban

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Education

Purdue University — ***West Lafayette, IN***
B.S. in Mechanical Engineering - Presidential Scholar | GPA: **3.2**/**4.0** — ***May 2027***

Technical Experience

Tesla
Cell Manufacturing Equipment/Process Design Engineering Intern — ***January 2026 - August 2026***

- Designed and deployed a shuttle-carried conveyor-cleaning tool with non-shedding foam to reduce grease-related cell contamination; cut maintenance downtime by **25%** and monthly cleaning intervention from two hours to 15 minutes.
- Redesigned winder protection to prevent shuttle damage across equipment generations; standardized a compression-fit tray through rapid prototyping and production trials, reducing damage by **20%** and enabling same-day replacement.
- Engineered and deployed a production jig to correct supplier-induced current-collector misalignment; integrated timed vibration control to center four tubes in **30 seconds**, with zero pickup faults over 60,000 cells in production trials.
- Improved inspection yield by **~10%** by qualifying a fleetwide fastener upgrade against vibration-induced loosening; developed an IMU-based accelerated test representing six months of runtime to reduce false rejection of good cells.
- Resolved laser-welding yield loss through root cause analysis, DOE, and JMP; optimized exhaust flow and validated weld quality with pull tests and dust checks, increasing downstream cell yield by **~10,000** cells/week.

Ford Motor Company
Product Development Intern - Forward Model — ***May 2025 - August 2025***

- Applied supplier-side 3L5Y and DFMEA analysis to address injector-tip cavitation and engine knock; updated failure modes for low-use/restart conditions and improved detection through additional extended fatigue tests.
- Improved the team DFMEA process by building its complete MagicDraw model for structured design failure analysis.
- Mapped real-world customer usage and refined test methods, improving in-field quality by **20%** during development.
- Redesigned operator instructions following plant Gemba visits to improve usability and UAW operator efficiency.

Purdue Electric Racing (Formula SAE EV)
Battery Senior Advisor — ***August 2023 - January 2026***

- Led battery-module redesign to improve structural integrity and increase specific energy by **30%** in Wh/kg; rerouted inertial loads around cell cans and verified the architecture through rule-based checks and hand calculations.
- Designed a thermal interface and spacer system to improve cooling, electrical isolation, and assembly fit; used tolerance analysis and material selection to control pad compression, improving battery performance by **10%**.
- Integrated cooling plates as structural walls within the pack architecture, reducing cooling-system weight by **11%** and pack width by 1.075 inches; used **FR4** barriers to maintain cell alignment and electrical insulation through assembly.
- Developed a cooling-plate validation rig to test thermal performance under representative cell compression; used Arduino temperature and flow logging to validate three plate designs within 1.2 degrees C of CFD predictions.
- Designed a waterproof battery enclosure that passed technical inspection without leaks, contributing to a top-10 finish.

Purdue Humanoid Robot Club
Quadruped Project Lead — ***August 2026 - Present***

- Lead mechanical design, motor selection, assembly, and team coordination for a custom quadruped robot.
- Led development of a prototype that walks unsupported and crouches while carrying an onboard NUC; used ROS/Gazebo simulation to support mechanical design reviews, motor integration, and system-level development.
- Direct a successor design targeting a jump four times body height, with planned onboard vision for platform detection.

Purdue MARS (Mechanisms and Robotic Systems) Lab
Undergraduate Researcher - NSF-funded project — ***August 2026 - Present***

- Developed an automated apple gripper/cutter to retain fruit during stem cutting using a custom servo-driven linkage.
- Integrated apple grasping, laser exposure, and stem cutting into one coordinated robot-arm operating sequence.
- Partnered with a software teammate to coordinate cutting using SAM/DINO outputs from Unitree Go2 EDU.

Tredence Inc.
Data Science Intern — ***May 2024 - August 2024***

- Identified sector trends using NumPy and Matplotlib in Python to inform retail decisions for clients.
- Developed regression models to analyze supply-chain data, reducing costs for key customers by **15%**.
- Used SQL for data cleaning, analysis, and visualization to translate findings into actionable client recommendations.

Skills

SolidWorks | 3DX | GD&T | Tolerance analysis | DFM | DFMEA | DOE | JMP | MATLAB | Python | SQL | Arduino | 3D printing